Euclidean Geometry In Mathematical Olympiads

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Problem 1.40 (Canada 1991/3)

Chapter 1 fgm fgmcon

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Kagebaka
3001 posts

Sep 30, 2018, 4:31 PM • 2 #5

Can someone explain why @above's proof works?

Maths_1729
390 posts

Mar 6, 2020, 9:23 AM • 1 #6

> realquarterb wrote:
> **Problem 1.40 (1991/3)** Let P be a point inside circle w. Consider the set of chords w that contain P. Prove That their midpoint all lie on a circle.

Let $M_1, M_2, M_3, ...M_n$ be the midpoints of all the chords passing through the point P and O be the centre of circle then as

$$\angle OM_nP = 90$$

for every n hence as

$$\angle OM_1P = \angle OM_2P = ... = \angle OM_nP = 90$$

which the angle made by same segment AP hence all the midpoints lie on a circle with diameter AP

mathlogician
1050 posts

Mar 18, 2020, 7:52 PM • 1 #7

solution

> We use the same definitions as in Post #2. Let M be the midpoint of chord AB. Clearly, Mw is perpendicular to AB, so $\angle AMw = 90$.Consider the circle with center O and radius OP. Since Pw is a diameter, any point K such that $\angle PKw = 90$ is on the circle, so M is on the circle.

Can someone check this please?

This post has been edited 1 time. Last edited by mathlogician, Mar 18, 2020, 7:52 PM

Kuroshio
72 posts

Mar 22, 2020, 3:08 PM • 1 #8

Consider the circle with diameter OP let's say ω.
Let AB be any chord passing through P with centre M.
$\angle OMP = 90^\circ \implies M$ lies on ω.The conclusion follows.

This post has been edited 2 times. Last edited by Kuroshio, Jul 18, 2020, 2:35 AM

Apr 21, 2020, 11:26 PM • 2 #9